Customer churn and feature engineering

A beginner's guide to logistic regression

A subscription app wants to identify customers who may cancel, so its team can offer timely support. Feature engineering turns activity records into useful predictors. Logistic regression combines those features to estimate a probability of cancellation.

Define the prediction

At the end of each month, predict whether an active customer will cancel during the next 30 days. The outcome is binary: churn = 1 for cancellation; churn = 0 otherwise. Each training row represents a customer at a monthly prediction date, paired with the outcome observed afterward.

Turn raw records into useful features

Raw data	Engineered feature	What it captures
Login timestamps	Days since last login	Recent disengagement
Monthly visit counts	Percentage change in visits	Increasing or declining usage
Subscription start date	Months as a customer	Customer tenure
Support ticket records	Number of unresolved tickets	Outstanding service problems
Payment records	Failed payments in past 30 days	Recent payment friction

Why a better feature can matter

Customer	Previous month	Latest month	Total visits	Usage change
A	20 visits	5 visits	25	−75%
B	5 visits	20 visits	25	+300%

Usage change = (latest − previous) ÷ previous × 100. Both customers have 25 visits, but their trends differ. A total-only model misses this distinction. Declining usage is a candidate churn signal to test, not proof that someone will cancel. If previous usage is zero, use a separate indicator or an absolute change instead.

Where logistic regression fits

The model learns a weight for each feature from historical examples. It adds the weighted features and an intercept, then applies the sigmoid function to produce a probability between 0 and 1. An illustrative prediction of 0.70 means an estimated 70% chance of cancellation within the next 30 days.

From customer records to model inputs

Fictional subscription app data. We make predictions at the end of April 30, 2026, using only information available then. March and April visit counts and the last login date come from the activity records. Churn is observed during the following 30 days.

ID	March visits Raw	April visits Raw	Last login Raw	Total visits Feature	Usage change Feature	Days since login Feature	Churn Label
C01	20	5	Apr 18	25	-75%	12	1
C02	5	20	Apr 30	25	+300%	0	0
C03	12	12	Apr 29	24	+0%	1	0
C04	16	4	Apr 15	20	-75%	15	1
C05	8	10	Apr 28	18	+25%	2	0
C06	30	15	Apr 27	45	-50%	3	0
C07	10	2	Apr 10	12	-80%	20	1
C08	4	8	Apr 30	12	+100%	0	0
C09	18	9	Apr 22	27	-50%	8	1
C10	6	3	Apr 26	9	-50%	4	0

How the features are calculated

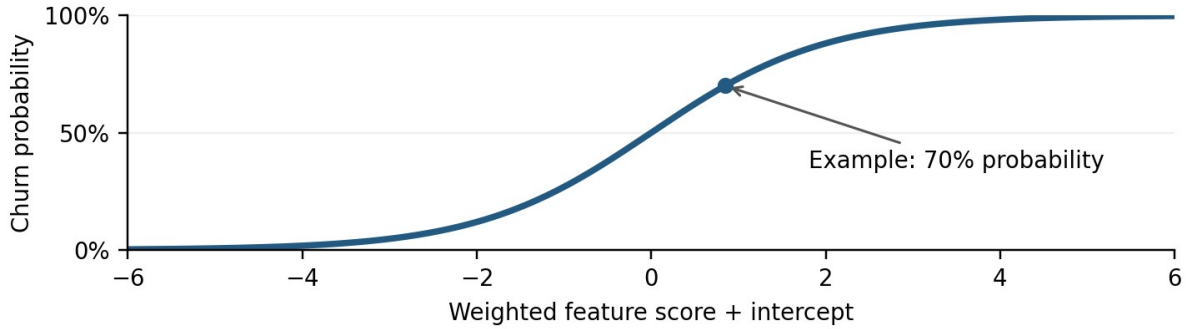
Total visits = March visits + April visits. Usage change = (April − March) ÷ March × 100. Days since login = April 30 − last login date. For C01: total = 25; usage change = −75%; days since login = 12.

C01 and C02 have the same total visits, but opposite usage trends. C06 and C09 both have a 50% decline, yet different outcomes. A feature can be informative without predicting every outcome correctly.

What goes into logistic regression

For this example, use the three engineered feature columns as inputs. Churn is the training label: 1 means canceled during the next 30 days; 0 means did not cancel. Customer IDs identify rows and are not model inputs. The raw columns are shown to make the calculations visible.

From a weighted score to a probability



The graph illustrates the sigmoid function, not a model fitted to these 10 customers. The marked 70% probability is illustrative; it is not a prediction for any row above.